

Typesetting Algorithms

1 The algorithmic Package

1.1 Conditional Statements and Looping

1. **if** Condition **then**
2. Carry out some operations.
3. **else if** Another Condition **then**
4. Carry out another operation.
5. **else**
6. Do basic operations.
7. **end if**
8. **for** $i = 0$ **to** 10 **do**
9. Carry out some operations.
10. **end for**
11. **while** Some condition holds **do**
12. Carry out some operations.
13. **end while**

1.2 Conditions and Logical Operators

Require: $x \neq 0$ and $n \geq 0$

Ensure: $x \neq 0$ and $n \geq 0$

1. **if** (value1 **and** value2) are equal **then**
2. Do something
3. **else if not** (value1 = value2) **then**
4. Do something else.
5. **else** {you can put a comment here}
6. **return true**
7. **end if**

2 The algorithm2e Package

Algorithm 2.1: Interval Restriction

Required: This is a required stuff.
Expected: This is the expected output range.
Data: $G = (X, U)$ such that G^{tc} is an order.
Result: $G' = (X, V)$ with $V \subseteq U$ such that G'^{tc} is an interval order.

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[1]: IntervalRest(parameters)
[2]: TestFunc(parameters) % This is a Matlab Style Comment
[3]:  $V \leftarrow U$ 
[4]:  $S \leftarrow \emptyset$ 
[5]: for  $x \in X$  do
[6]:    $NbSuccIns(x) \leftarrow 0$ 
[7]:    $NbPredInMin(x) \leftarrow 0$ 
[8]:    $NbPredNotInMin(x) \leftarrow |ImPred(x)|$ 
[9]: for  $x \in X$  do /* This is a C comment */
[10]:   if  $NbPredInMin(x) = 0$  and  $NbPredNotInMin(x) = 0$  then
[11]:      $AppendToMin(x)$ 
[12]: while  $S \neq \emptyset$  do
[13]:   remove  $x$  from the list of  $T$  of maximal index.
[14]:   while  $|S \cap ImSucc(x)| \neq |S|$  do
[15]:     for  $y \in S - ImSucc(x)$  do
[16]:       { remove from  $V$  all the arcs  $zy$ : }
[17]:       for  $z \in ImPred(y) \cap Min$  do
[18]:         remove the arc  $zy$  from  $V$ 
[19]:          $NbSuccIns(z) \leftarrow NbSuccIns(z) - 1$ 
[20]:         move  $z$  in  $T$  to the list preceding its present list
[21]:         {i.e. If  $z \in T[k]$ , move  $z$  from  $T[k]$  to  $T[k - 1]$ }
[22]:        $NbPredInMin(y) \leftarrow 0$ 
[23]:        $NbPredNotInMin(y) \leftarrow 0$ 
[24]:        $S \leftarrow S - \{y\}$ 
[25]:        $AppendToMin(y)$ 
[26]:    $RemoveFromMin(x)$ 

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